

PlasticWatch

Plastic Waste Intelligence Command Center

An explainable AI-GIS workflow that turns image reports into verified cleanup priorities.

DEMO / SIMULATION

Synthetic reports are clearly labelled. Human teams verify every hotspot before a cleanup decision.



**REPORT
AI EVIDENCE
HOTSPOT
VERIFY
CLEANUP**

THE CHALLENGE

Plastic waste reports rarely become actionable work orders

PS-08 asks for detection, geospatial clustering, prioritisation, verification, and cleanup planning.

01 Scattered reports

Images and complaints sit in separate channels.

02 Weak evidence

A single image can show plastic, algae, glare, or mixed waste.

03 Water exposure

Sites near drains can carry waste into water bodies.

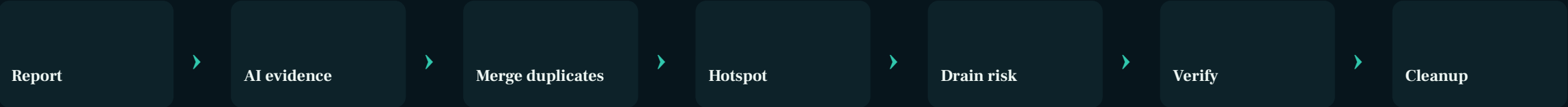
04 Reactive response

Teams lack a ranked, auditable verification queue.

PlasticWatch creates one evidence trail from report to verified field action.

Evidence-to-action workflow

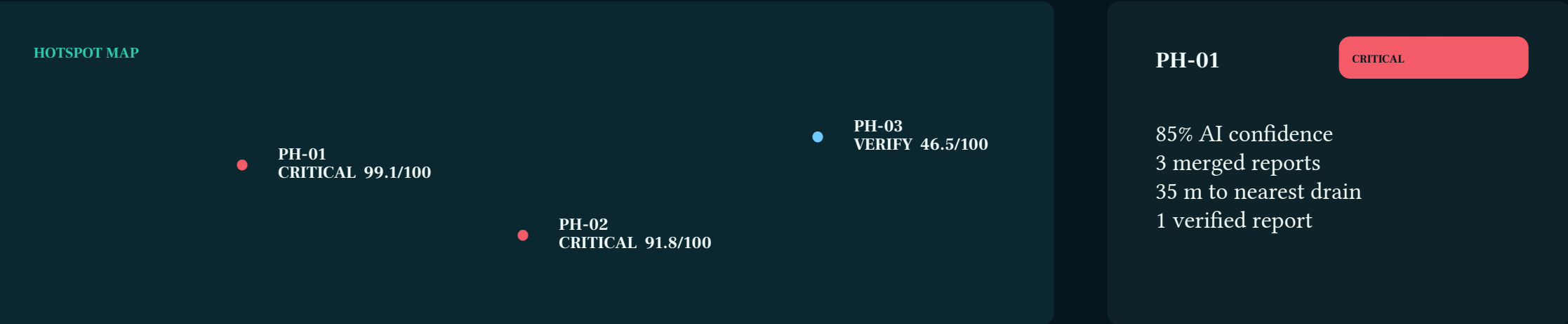
The prototype keeps AI advisory. Ward teams make verification and cleanup decisions.



Before / after evidence closes the loop and becomes training feedback.

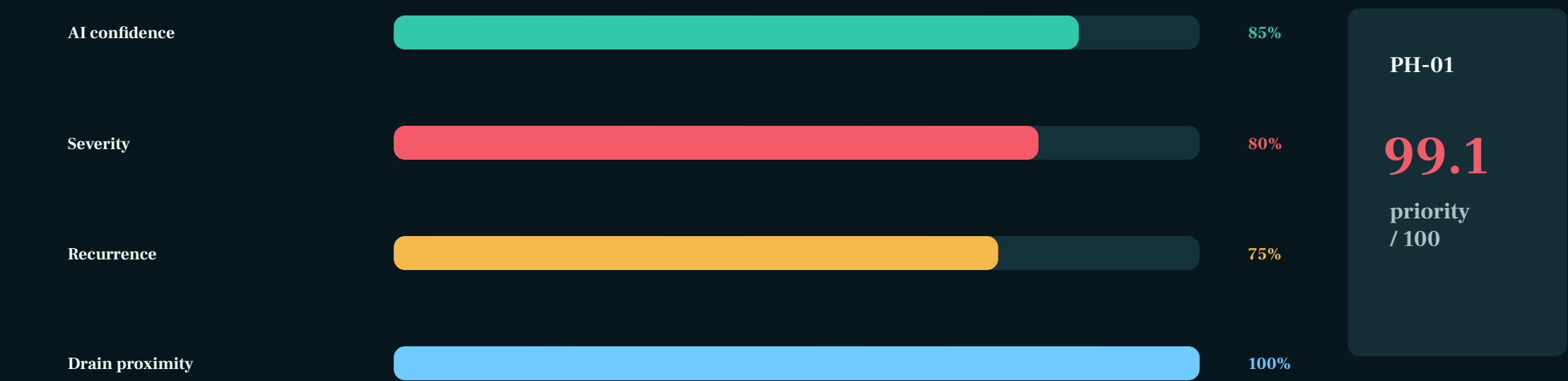
A Pune municipal pilot turns six reports into three hotspots

All values below come from the current transparent replay dataset.



PH-01 ranks first because the evidence and water exposure are both strong

Priority uses the existing scoring components rather than a black-box recommendation.



Why this score? Expandable details in the dashboard show report evidence and the formula.

TACO provides labelled waste images for the next detector stage

TACO is training data, not a trained detector.

1,500

images

4,784

labelled objects

60

categories

4

project classes

Training path

TACO labels



4-class map



Fine-tune detector



Image upload



Plastic confidence

Classes: plastic bottle · bag / wrapper · mixed plastic · uncertain / non-plastic

Verification protects teams from false positives and creates before / after evidence

The queue recommends inspection. It does not trigger autonomous enforcement or assign responsibility.

01

PH-01

VERIFY NOW

99.1 / 100

3 reports · 35 m to drain

Collect photo + GPS
✓ Confirm material type
Approve cleanup if verified

02

PH-02

VERIFY NOW

91.8 / 100

2 reports · 20 m to drain

Collect photo + GPS
✓ Confirm material type
Approve cleanup if verified

03

PH-03

REVIEW

46.5 / 100

1 report · 210 m to drain

Collect photo + GPS
✓ Confirm material type
Approve cleanup if verified

Outcome record: reported → verified / rejected → cleaned → before / after evidence

One common report schema lets PlasticWatch expand from a demo to ward operations

The system separates evidence, prioritisation, verification, and cleanup records.

Inputs

Citizen image
Ward-worker report
Street-camera frame

Decision support

Detector confidence
Duplicate clustering
Drain-risk score

Field loop

Verification result
Before / after evidence
Model feedback

Safeguard

AI confidence is evidence, not final truth. Human teams verify material type before a cleanup decision or any public claim.

A 90-second walkthrough of the working prototype

Use the dashboard, not slides, for the main product proof.

0-10 s	Problem and mode	Open the header. Point out PS-08 and the DEMO / SIMULATION label.
10-35 s	Hotspot map	Select PH-01. Explain duplicate merging, score, and drain distance.
35-55 s	Evidence	Open Evidence & AI. Show confidence and verification state.
55-75 s	Field queue	Open Verification Queue. Explain the order and human decision gate.
75-90 s	TACO readiness	Show the TACO status. State that training data is ready, while the detector remains the next phase.